

Wireless Communications Using MATLAB

14th April 2024

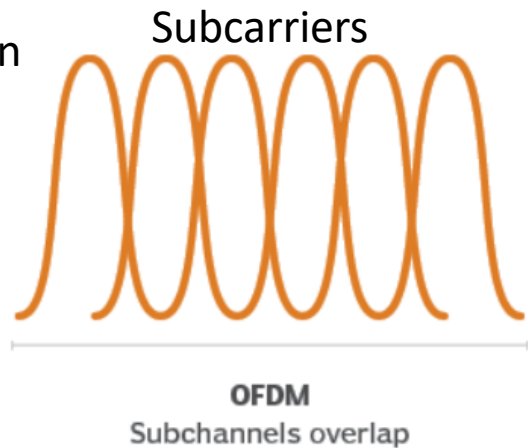
Dr. Saif Ali Jasim Al Grait

Orthogonal frequency-division multiplexing is a method of data transmission where a single information stream is split among several closely spaced narrowband subchannel frequencies instead of a single Wideband channel frequency

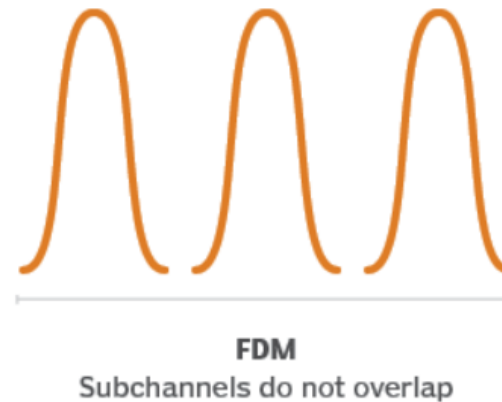
How orthogonal frequency-division multiplexing (OFDM), standard frequency-division multiplexing (FDM) and single-channel wireless transmission compare.

OFDM Used in

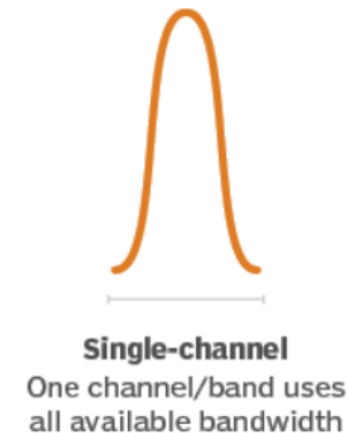
- WiFi
- LTE
- 5G



- No guard band
- Higher data rate than FDM for same BW
(spectrally efficient)

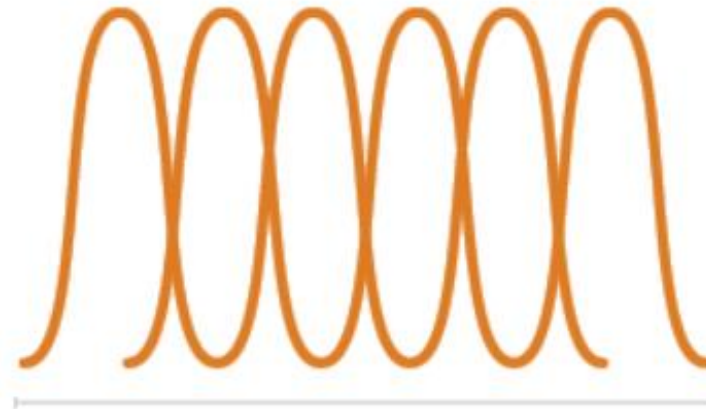


Allows multiple users
To share one link
by dividing BW between
Users, a guard band is needed



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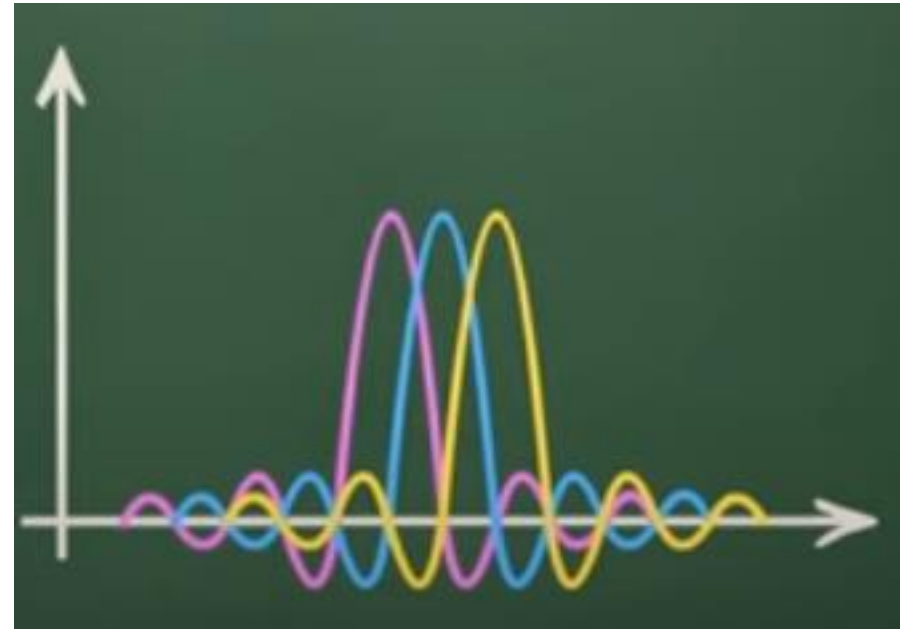
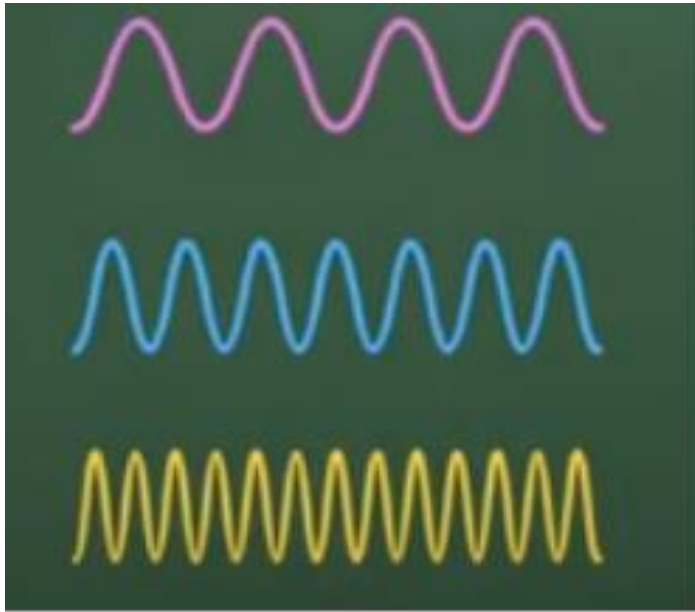
Why there is no inference between these subchannels?



OFDM
Subchannels overlap

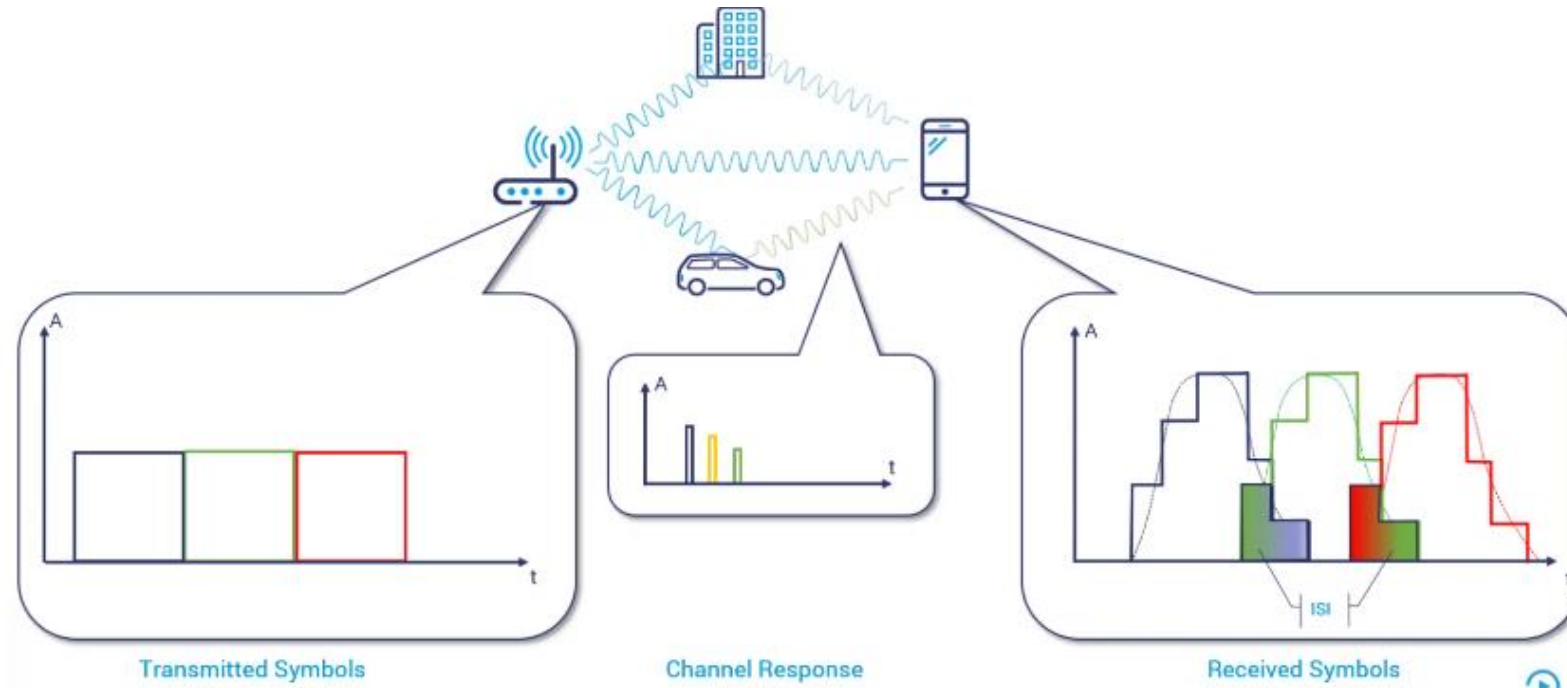
OFDM combines these signals to be orthogonal with each other

- Orthogonal word means two or multiple objects act independently
- In the OFDM case, OFDM operates without dependence or interference with one another



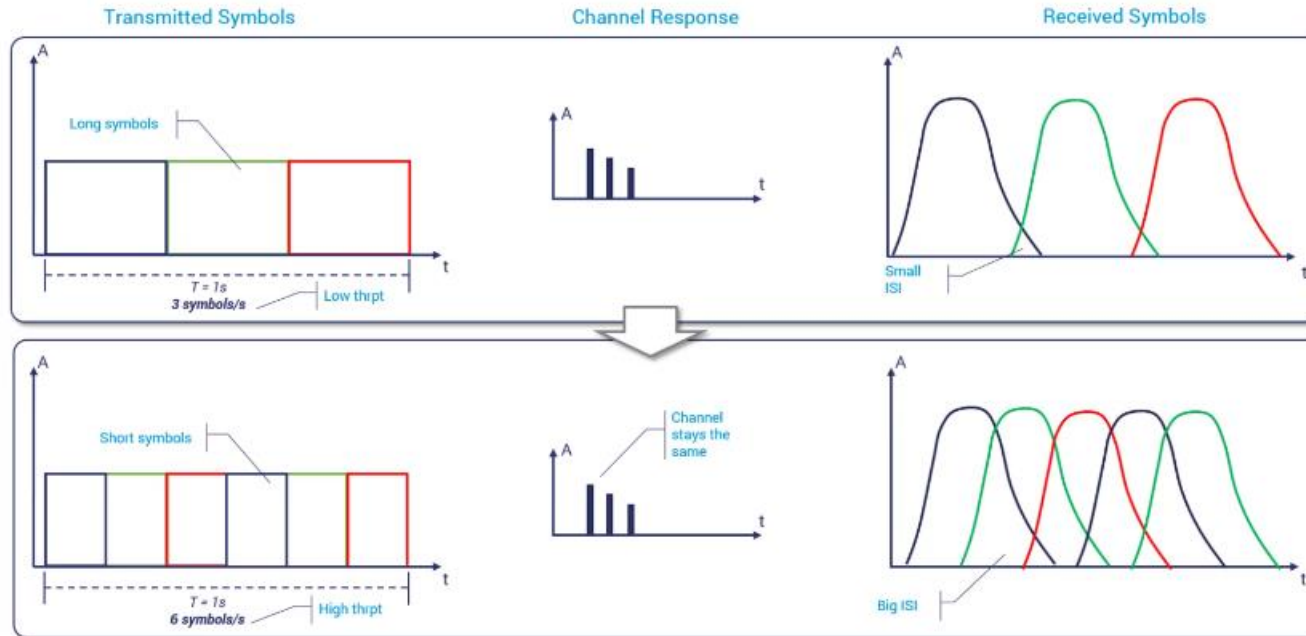
- These signals are orthogonal because when a signal reaches the peak its two neighbors are zero or at null value
- At the receiving side the de-multiplexer will separate them based on this orthogonal feature

Multipath channel recap



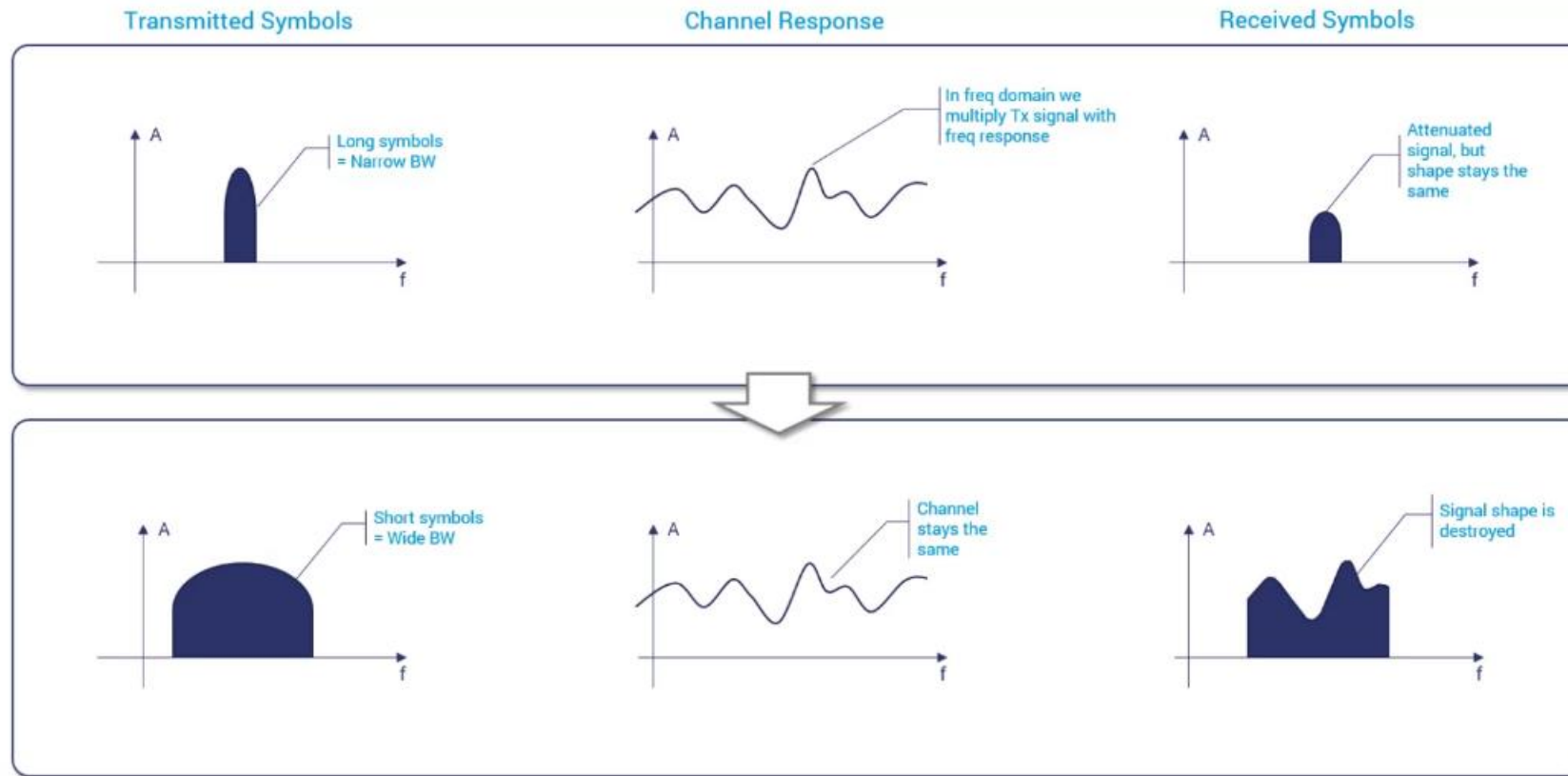
- “Transmit a single rectangle symbol, it shows up at the receiver as a combination of (in this case 3 copies) that are differently delayed and attenuated
- When there are multiple consecutive symbols, one right next to another – this results in interference from the previous symbol to the next – Inter-Symbol Interference (ISI)”

Inter-Symbol Interference



- “Let’s assume, we have 3 symbols transmitted within 1 second (3 sym/sec), each carrying, for simplicity, a single bit. (Otherwise each symbol can carry multiple bits, as in the case of QPSK/QAM mapping)”

Long symbols result in narrow Bandwidth ($BW = \sim 1/T$)



How orthogonal frequency-division multiplexing works

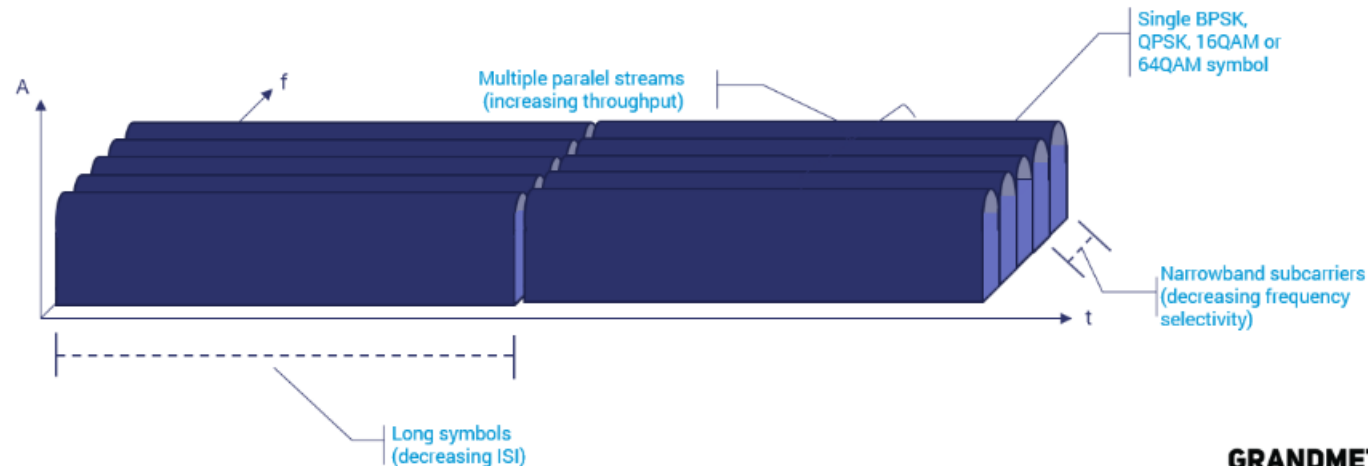
- Long symbols mean a low impact on the ISI
- High throughput is needed, which is impossible through single carrier modulation, where data symbols (BPSK, QPSK, and QAM) transmitted one after another have to be very short because of the increasing impact of the channel.

So... we want long signals – because channel impact is low (low ISI, not destroyed shape) BUT we also want high throughput in the same time!



This is possible via OFDM...

... where the usable bandwidth is divided into a large number of smaller bandwidths, or “subcarriers” (SCs). The high-speed information is divided onto multiple lower-speed signals, transmitted simultaneously on different frequencies in parallel.



Benefits of the orthogonal frequency-division multiplexing

“This is where OFDM comes as a solution. It, simply speaking, combines all requirements so that we get:

- Long symbols – decreasing ISI,
- Narrow BWs – avoiding frequency selectivity,
- High throughput”

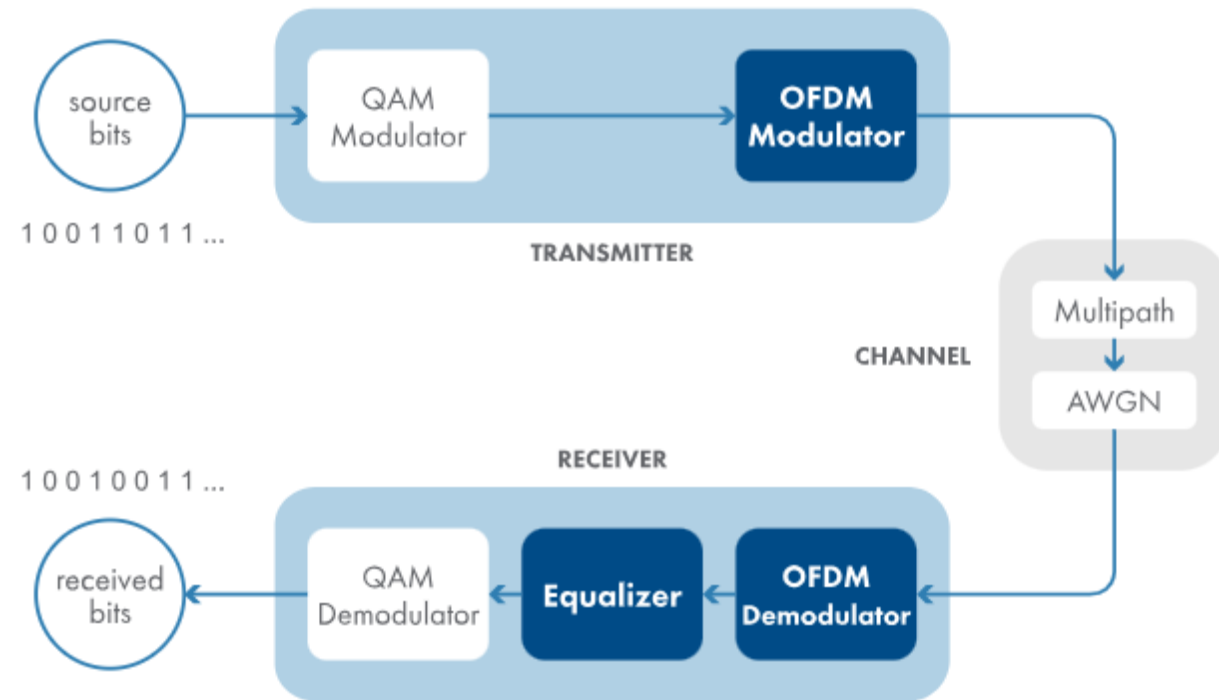


Frequency Domain Equalization

In this activity, you'll simulate an OFDM system, which includes a cyclic prefix and a frequency domain equalizer.

As with the IFFT, the OFDM modulator sends one 16-QAM symbol on each subcarrier.

- It then adds a cyclic prefix to create a single OFDM symbol, which is sent over the channel.
- In the receiver, the OFDM demodulator removes the cyclic prefix and performs an FFT. You then equalize the output in the frequency domain via division.



When designing a practical OFDM system, the number of subcarriers (the FFT size) is an important design parameter

The simulation in this activity is designed around the number of subcarriers—one 16-QAM symbol is assigned to each subcarrier, creating a single OFDM symbol. The number of source bits to generate is calculated by multiplying the number of subcarriers by the bits per 16-QAM symbol.

TASK

Define the number of subcarriers to be 8192 and store it in the variable numCarr. Then calculate the number of source bits to generate and store it in the variable numBits.

After setting the number of source bits, the script creates the source bit sequence and modulates it using 16-QAM

The next step is to perform OFDM modulation, including a cyclic prefix.

The `ofdmmod` and `ofdmdemod` functions manage the cyclic prefix for you. All you have to do is specify the length of the cyclic prefix.

The cyclic prefix must be at least as long as the impulse response of the channel. But, it should also be reasonably short because it contains duplicate data that will be discarded. In this simulation, the multipath channel impulse response length is 17, so use a value slightly larger than that for the cyclic prefix length.

TASK

Define the cyclic prefix length to be 32 and store it in the variable `cycPrefLen`.

Use the `ofdmmod` function to perform OFDM modulation, specifying the number of subcarriers and the cyclic prefix length

```
sigOut = ofdmmod(sig,nCarr,cpLen)
```

TASK

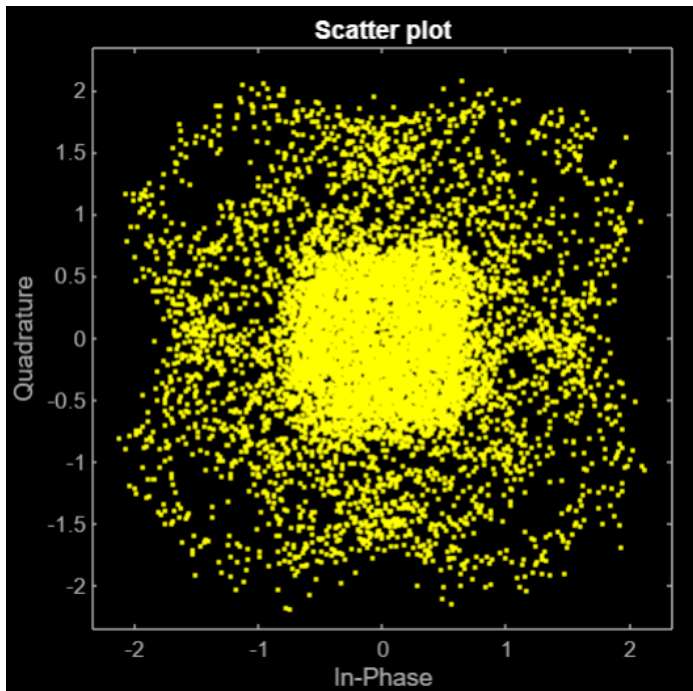
Perform OFDM modulation on the signal `qamModOut`. Specify the number of subcarriers as `numCarr` and the cyclic prefix length as `cycPrefLen`.

Name the output signal `ofdmModOut`.

Next, the OFDM modulated signal is sent over the channel to the receiver

In the receiver, perform OFDM demodulation using the `ofdmmodemod` function, again specifying the number of subcarriers and the cyclic prefix length.

```
sigOut = ofdmmodemod(sig,nCarr,cpLen)
```



TASK

Perform OFDM demodulation on the signal `chanOut`. Specify the number of subcarriers as `numCarr` and the cyclic prefix length as `cycPrefLen`.

Name the output signal `ofdmDemodOut`.

Create a scatter plot of the OFDM demodulator output.

References

[1] <https://www.youtube.com/watch?v=p1PbJaraL3g>